

Assignment

Creating Dashboard with Visualization Tool

1) What is Power BI and how does it differ from Excel?

Ans - Microsoft Power BI is a tool used to create interactive dashboards and visualize large datasets, while Microsoft Excel is mainly used for data analysis, calculations, and basic reporting.

Power BI handles large data efficiently and provides real-time insights, whereas Excel is better for detailed data manipulation.

Example: A company may use Excel to clean and prepare sales data, and then use Power BI to build a dashboard showing monthly sales trends and KPIs.

2) Explain the concept of data modeling in Power BI.

Ans - Data modeling in Microsoft Power BI is the process of creating relationships between different tables to organize data for efficient analysis.

It typically involves structuring data into fact and dimension tables (star schema). A good data model improves performance and ensures accurate reporting.

Example: Linking a Sales table with Customer and Product tables helps analyze sales by customer or product easily.

3) What are the different types of connections available in Power BI?

Ans - In Microsoft Power BI, there are three main types of data connections:

- **Import** → Data is loaded into Power BI, giving fast performance.
- **Direct Query** → Data is not stored; it is fetched in real-time from the source.
- **Live Connection** → Connects directly to an existing data model (like Analysis Services).

Example: If you want real-time sales data from a database, you would use Direct Query.

4) How do you handle data transformation in Power BI?

Ans - Data transformation in Microsoft Power BI is handled using the Power Query Editor, where we clean and shape the data before analysis.

It includes tasks like removing duplicates, handling missing values, changing data types, and merging tables.

This process ensures the data is accurate and ready for reporting.

Example: Converting a text column into date format and removing blank rows before creating a dashboard.

5) What is DAX (Data Analysis Expressions) and why is it important in Power BI?

Ans - DAX (Data Analysis Expressions) in Microsoft Power BI is a formula language used to create calculations like measures and calculated columns.

It helps in performing advanced analysis such as totals, averages, and percentage calculations.

DAX is important because it makes reports dynamic and allows deeper insights from data.

Example: Creating a measure like Total Sales = SUM(Sales[Amount]) to calculate overall sales.

6) Can you explain the difference between calculated columns and measures in Power BI?

Ans - In Microsoft Power BI, a calculated column is created row-by-row and stored in the table, increasing the data size.

A measure is calculated dynamically based on filters and is not stored, making it more efficient.

Calculated columns are used for static calculations, while measures are used for aggregations like sum or average.

Example: A column for “Profit = Sales – Cost” is a calculated column, while “Total Profit = SUM(Profit)” is a measure.

7) How do you handle relationships between tables in Power BI?

Ans – In Microsoft Power BI, relationships between tables are handled by connecting them using common columns (keys), such as ID fields.

These relationships are usually defined as one-to-many or many-to-one in the data model view.

Proper relationships ensure accurate filtering and data flow between tables.

Example: Linking a Customer table with a Sales table using Customer ID allows you to analyze sales by each customer.

8) What is the purpose of a Power BI Gateway?

Ans - The Power BI Gateway in Microsoft Power BI is used to securely connect on-premises data sources to the Power BI Service in the cloud.

It allows data to be refreshed automatically without moving the data permanently to the cloud.

This ensures real-time or scheduled updates for reports and dashboards.

Example: Connecting a local SQL Server database to Power BI Service so reports update daily.

9) How can you schedule data refresh in Power BI Service?

Ans - In Microsoft Power BI Service, you can schedule data refresh by publishing your report, then going to Dataset Settings → Scheduled Refresh.

Enable the refresh option, set the frequency (daily/weekly), and provide data source credentials.

If the data is on-premises, you also need to configure a Gateway.

Example: Setting a daily refresh at 8 AM so your sales dashboard always shows the latest data.

10) Explain the concept of row-level security in Power BI.

Ans - Row-level security (RLS) in Microsoft Power BI is used to restrict data access for different users based on their roles.

It ensures that users can only see the data relevant to them by applying filters at the row level.

This improves data security and privacy in reports.

Example: A sales manager can only view data for their own region, not the entire company.

11) What is the Power BI Desktop and how does it differ from Power BI Service?

Ans - Microsoft Power BI Desktop is a local application used to create reports, transform data, and build data models.

Microsoft Power BI Service is an online platform used to publish, share, and collaborate on reports and dashboards.

Desktop is mainly for development, while Service is for sharing and accessing reports from anywhere.

Example: You create a sales dashboard in Desktop and then publish it to Service for your team to view online.

12) Explain the concept of Direct Query in Power BI ?

Ans - DirectQuery in Microsoft Power BI is a connection mode where data is not stored in Power BI, but fetched in real-time from the data source.

Every time you interact with the report, Power BI sends a query to the source and shows updated results.

It is useful when working with large or frequently changing data.

Example: Connecting to a live SQL database to view real-time sales updates without importing the data.

13) What are Power BI templates and how are they useful?

Ans - Power BI templates in Microsoft Power BI are reusable report files (.pbix) that contain the report layout, visuals, and data model but not the actual data.

They help save time by allowing you to reuse the same structure with different datasets.

Templates ensure consistency across reports and make sharing easier.

Example: Creating a sales dashboard template and using it for different regions by just connecting new data.

14) How do you handle incremental data refresh in Power BI.

Ans - Incremental refresh in Microsoft Power BI is used to load only new or updated data instead of refreshing the entire dataset.

It is configured using date/time filters (RangeStart and RangeEnd) in Power Query and then set up in the model settings.

This improves performance and reduces refresh time for large datasets.

Example: Refreshing only the last 7 days of sales data instead of reloading the full historical data every time.

15) What is the role of Power Query in Power BI?

Ans - Power Query in Microsoft Power BI is used for data extraction, cleaning, and transformation (ETL process) before analysis.

It helps in tasks like removing duplicates, handling missing values, changing data types, and merging tables.

This ensures the data is clean and structured for accurate reporting.

Example: Combining multiple Excel files and cleaning the data before creating a dashboard.

16) Explain the difference between calculated columns and calculated tables in Power BI.

Ans - In Microsoft Power BI, a calculated column is added to an existing table and computes values row-by-row, and it is stored in the model.

A calculated table is a completely new table created using DAX formulas based on existing data.

Calculated columns are used for adding new fields, while calculated tables are used for creating new datasets or summaries.

Example: Adding a “Profit” column is a calculated column, while creating a “Top Customers” table using a DAX filter is a calculated table.

17) How do you create custom visuals in Power BI?

Ans - In Microsoft Power BI, custom visuals can be created by either importing them from the Microsoft AppSource marketplace or building your own using the Power BI Visuals SDK.

You can add a custom visual by going to the Visualizations pane → “Get more visuals” → import or select from marketplace.

This allows you to use advanced and unique charts beyond default visuals.

Example: Adding a custom funnel or KPI visual to better represent sales performance.

18) What are the best practices for optimizing performance in Power BI?

Ans - In Microsoft Power BI, performance can be optimized by using a star schema, reducing unnecessary columns, and keeping the data model simple.

Prefer measures over calculated columns and write efficient DAX formulas.

Also, limit the number of visuals on a report page and use filters to reduce data load.

Example: Removing unused columns and using aggregated tables to make dashboards load faster.

19) How can you integrate Power BI with other Microsoft products like Azure and Office 365?

Ans - Microsoft Power BI integrates with Microsoft products like Microsoft Azure, Microsoft Excel, Microsoft SharePoint, and Microsoft Teams to improve data analysis and sharing.

It can connect to Azure services like SQL Database and Data Lake for cloud-based data storage and real-time analytics.

With Office 365 tools, reports can be shared, embedded, and collaborated on easily across teams.

Example: Connecting Power BI to Azure SQL Database and sharing dashboards directly in Teams for business decision-making.

20) Explain the concept of aggregations in Power BI.

Ans - In Microsoft Power BI, aggregations refer to summarizing data using functions like SUM, COUNT, AVERAGE, MIN, and MAX.

They are used to create summarized tables that improve query performance, especially with large datasets.

Power BI can use pre-aggregated data to reduce load time and speed up reports.

Example: Instead of calculating total sales from millions of rows, using a pre-aggregated monthly sales table for faster results.

21) How do you handle error handling and data quality in Power BI?

Ans - In Microsoft Power BI, error handling and data quality are managed mainly using Power Query Editor during data transformation.

You can detect and fix errors by replacing error values, removing invalid rows, handling nulls, and setting correct data types.

Validations and filters are applied to ensure clean and consistent data before loading it into the model.

Example: Removing rows with errors or replacing null values with defaults before creating reports to ensure accurate analysis.

22) What is the purpose of Power BI Embedded and when would you use it?

Ans - Power BI Embedded in Microsoft Power BI is used to integrate Power BI reports and dashboards directly into custom applications or websites.

It allows developers to provide interactive analytics to users without requiring them to open Power BI separately.

This is useful for delivering data insights within business apps or customer-facing platforms.

Example: Embedding a sales dashboard inside a company's web application for clients to view their performance.